

Grouping in pairs



1 Factorise:

a $5(a + b) + v(a + b)$

c $5y(4w + 3x) - z(4w + 3x)$

e $3f(g + h) + (g + h)^2$

g $(2c - d)(c + 5d) - 3(d - 2c)$

b $x(y - z) - w(y - z)$

d $2y(2x^2 + 3z) - (2x^2 + 3z)$

f $8y(y - 4) + 3(4 - y)$

h $3x^2(x + 4y) - 5y(x + 4y)$

2 Factorise:

a $8x + xz - 16y - 2yz$

c $7xy + wx + 7yz + wz$

e $x^2 - 3x + 8x - 24$

g $a^3 + 8a^2 + a + 8$

i $4x + 24yz + 32xy + 3z$

b $24 + 3y + 8x + xy$

d $x^2 + 2x + 5x + 10$

f $2mp + 6 + 3p + 4m$

h $3pq^2 - 11ypq + 3rsq - 11yrs$

j $2x + 18yz + 12xy + 3z$

Perfect squares

3 Factorise:

a $q^2 + 2qt + t^2$

c $b^2 - 2br + r^2$

e $c^2 - 4c + 4$

g $49 - 2p + p^2$

i $64r^2 + 48r + 9$

b $u^2 - 2uq + q^2$

d $x^2 + 6x + 9$

f $64 + 2e + e^2$

h $16w^2 - 40w + 25$

j $\frac{1}{4} - 3d + 9d^2$

4 Factorise:

a $x^4 + 8x^2 + 16$

b $a^4 - 18a^2 + 81$

5 A square has area $t^2 + 14t + 49$.

a Find the side length of the square.

b Find the perimeter of the square.



Difference of two squares

6 One linear factor of $x^2 - 16$ is $x - 4$. Find the other factor.

7 What can we multiply $z + 7$ by to give the difference of two squares?

8 Factorise:

a $x^2 - y^2$

b $n^2 - 25$

c $v^2 - 1$

d $121 - v^2$

e $x^2 - \frac{1}{4}$

f $x^2 - \frac{25}{121}$

g $16 - 9y^2$

h $x^2y^2 - 49$

i $25m^2 - 49$

j $81x^2 - 16y^2$

k $7x^2 - 63$

l $45t^2 - 20$

 9 Using the difference of two squares, evaluate the following:

a $65^2 - 63^2$

b $99^2 - 97^2$

c $45^2 - 42^2$

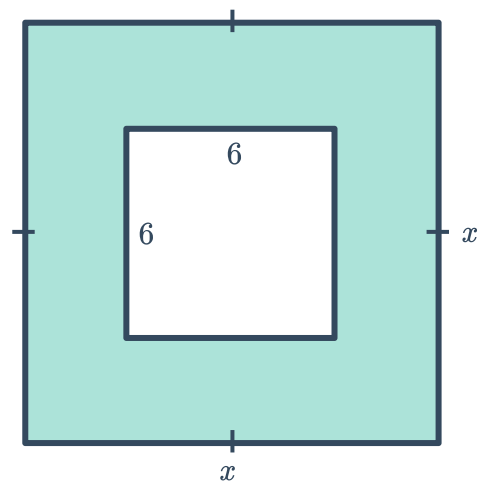
d $58^2 - 52^2$

 10 Find the value of the following by rewriting in the form $(x + y)(x - y)$:

a 504×49

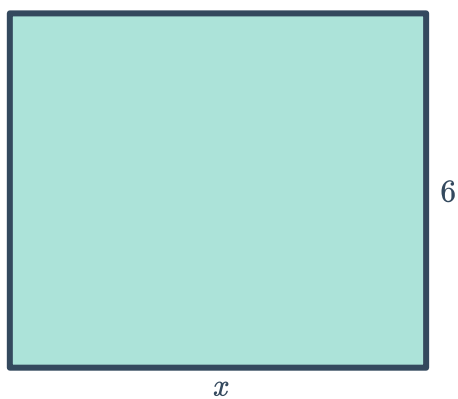
b 304×296

11 Consider the shaded area shown:

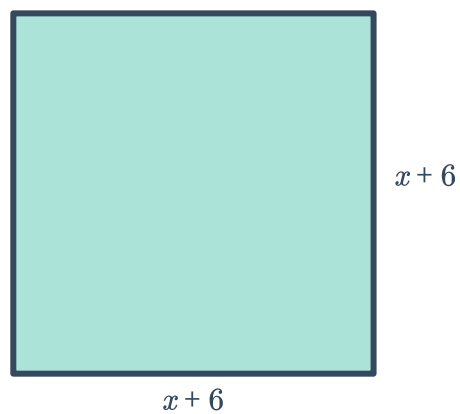


- a Find the area of the shaded region.
- b State whether the following rectangles have the same area as the shaded region.

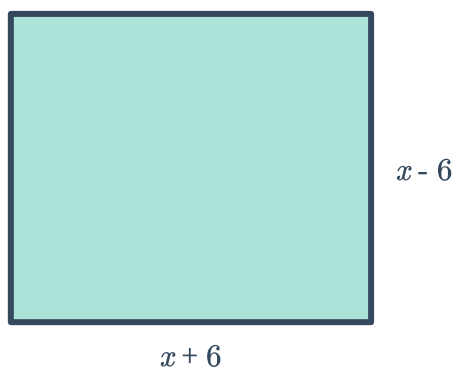
i



ii



iii





Mixed factorisations

12 Factorise:

a $3x^2y - 6y^4 + 9x^2y^3$

c $121c^2 - 49d^2$

e $a^2 + 2ab + b^2 - c^2$

g $(b + 4)^2 - (3 - b)^2$

i $8w^2 - 32wt^2 + 3wt - 12t^3$

k $48df^2 - 3dh^2$

m $18j^2 + 60jk + 50k^2$

b $16x^2y^2 + 24xyz + 9z^2$

d $3x^2y + x^2z + 3y^2 + yz$

f $8a^2b^4 + 12bc^2$

h $14h^3p^2r + 21h^5p^2 + 49hp^4$

j $16q^4 + 24q^2r + 9r^2$

l $128t^3 - 18s^4t$

n $x^2y + 7x^2 - y - 7$

13 Factorise $x^4 - 1$.